

The Language-Only Hypothesis: Emergent Capabilities in Large Language Models Are Properties of Natural Language Structure, Not of Neural Networks or Scale

First Step and Pre-registered Ablation

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Abstract

A strong form of the “emergent abilities” explanation holds that qualitatively new capabilities appear when language models are scaled past certain universal thresholds of parameters and training data. We propose the far stronger Language-Only Hypothesis: these capabilities are not emergent from scale or architecture at all, but are inevitable statistical consequences of the deep, multi-scale structure intrinsic to natural language. As a first, rigorous test, we train decoder-only Transformers of identical architecture (125 M and 350 M parameters) from scratch — one on English web text, one on French web text — holding every technological variable constant. We pre-register the prediction that French-trained models will reach multiple capability thresholds dramatically earlier than English-trained models of up to $3\times$ larger scale. If confirmed, the result falsifies scale as the primary driver of emergence and provides the first direct support for the Language-Only Hypothesis. Training is ongoing; full results will be reported in a follow-up manuscript.

1 Introduction

The past five years have seen the rapid acceptance of a single explanatory framework for the capabilities of large language models: that qualitatively new behaviors “emerge” when models are scaled past certain thresholds of parameters and training data Wei et al. [2022], Hoffmann et al. [2022], Schaeffer et al. [2024]. This narrative has become the organizing principle of the field, justifying ever-larger models and the accompanying expenditure of energy and capital.

Yet the same scaling recipe — decoder-only Transformers trained with next-token prediction — has been applied to numerous other domains with dramatically different outcomes. Vision models trained on billions of images Radford et al. [2021], Chen et al. [2024] still fail systematically at long-horizon planning and compositional reasoning. Audio models trained on billions of seconds of speech Baevski et al. [2023], Borsos et al. [2022] do not spontaneously develop theory-of-mind or counterfactual reasoning. Protein models trained on the entire known sequence database Lin et al. [2023], Rives et al. [2021] do not hallucinate coherent multi-paragraph scientific papers. Robotics models trained on millions of real-world trajectories Brohan et al. [2023], Kim et al. [2024] remain brittle outside their training distribution. In every non-linguistic domain, scaling alone has failed to produce the kind of sharp, broad, unpredictable capability jumps routinely observed in language models.

2 The Language-Only Hypothesis

We stake the following strong claim, which we intend to test incrementally across multiple experiments:

All capabilities currently described as “emergent” in large language models are predictable statistical consequences of deep, multi-scale regularities intrinsic to natural language corpora, and are not caused by neural-network architecture, parameter count, or training compute.

The neural network and optimiser serve as instrumentation — replaceable measuring devices that reveal structure already present in the data. Change the instrument and the revealed capabilities change little. Change the language and the revealed capabilities change dramatically.

This paper reports the first controlled ablation toward testing the hypothesis: we vary only the training language (English vs. French) while holding architecture, scale, optimiser, tokenizer, and token volume identical. Future work will test replaceability of the architecture and systematic destruction of linguistic structure.

Recent analysis has confirmed that natural language exhibits robust fractal scaling across many languages [Loff et al., 2023], providing empirical grounding for the existence of the deep structure we posit.

If the hypothesis is correct:

1. A 125 M-parameter model trained on French will reach any fixed capability threshold using substantially fewer tokens than an identical model trained on English.
2. At convergence, the 125 M French model will outperform the 350 M English model on multiple independent capability clusters.
3. These divergences will be quantitatively predictable from purely linguistic corpus statistics (e.g., mean recursion depth, clitic-to-content ratio, liaison density).

A single robust counterexample falsifies the hypothesis.

This experiment is the cleanest possible ablation: every technological variable is held constant; only the language itself varies. Training is underway.

3 Pre-registered Experiment

We train four decoder-only Transformer language models from random initialization:

- 125 M parameters on approximately 20 billion tokens of English web text (C4 English)
- 125 M parameters on approximately 20 billion tokens of French web text (C4 French)
- 350 M parameters on the same English corpus (control)
- 350 M parameters on the same French corpus (control)

3.1 Controlled variables (identical across all four runs)

- Architecture: standard decoder-only Transformer with rotary positional embeddings, SwiGLU feed-forward layers, and RMSNorm

- Hyper-parameters: 1024-token context, cosine learning-rate schedule with 100 warmup steps, peak learning rate 6×10^{-4} , AdamW $\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay 0.1
- Batch size: 0.5 M tokens per gradient step (achieved via gradient accumulation when necessary)
- Optimizer state and random seeds: identical initialization scheme and seed sequence across languages
- Tokenizer: single 50,257-token BPE vocabulary trained jointly on 1 M English + 1 M French documents from the respective corpora (exact same merges for all runs)
- Training duration: continued until validation perplexity plateaus or 20 B tokens are processed, whichever comes first
- Hardware and software: MLX framework on Apple-silicon, mixed-precision (bfloat16), no model parallelism

The ****only variable**** that differs between paired runs is the natural language used in the training corpus.

Training began on 30 November 2025 and is ongoing. All code, tokenizer, data-processing pipeline, and training logs are publicly available at <https://github.com/adamzwasserman/fractal-language>. Checkpoints will be released immediately upon completion of each run.

4 Pre-registered Predictions

We pre-register the following three predictions before any model has completed training.

1. **Token-efficiency threshold prediction.** Despite French’s substantially higher morphological density (which already confers a 40% token-level information advantage), the 125 M French model will reach any fixed capability threshold ($\geq 70\%$ on held-out benchmarks) using **$\geq 60\%$ of the raw tokens** required by the identical 125 M English model — a net advantage that cannot be explained by tokenizer effects alone.
2. **Parameter-efficiency crossover prediction.** At convergence, the 125 M French model will outperform the 350 M English model on at least three of the following four capability clusters:
 - syntactic recursion depth ≥ 7 on centre-embedded constructions
 - 8-shot in-context learning on symbolic reasoning tasks
 - stable persona coherence across 20-turn dialogues
 - susceptibility to indirect suggestion (embedded-command acceptance rate)
3. **Predictability-from-structure prediction.** The observed token-efficiency ratio (French tokens / English tokens) for each capability will correlate strongly ($r > 0.8$) with independently measured linguistic fractal and recursive statistics of the two corpora [Loff et al., 2023] (Hurst exponent, mean maximum embedding depth, clitic-to-content ratio, liaison density).

A single confirmed prediction with statistical significance falsifies the claim that scale is the primary driver of emergent capabilities.

All benchmarks, prompting templates, scoring scripts, and random seeds are fixed as of 30 November 2025 and will be released with this manuscript.

5 Discussion

A positive result would imply that the field has been looking in the wrong place for the last five years. The path forward is not larger models, but deeper understanding of the statistical physics of human language.

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